365 Real Estate — Residential Valuation

Project report draft | Linear Regression

Business question

Estimate residential property sale price in AED using information a valuer could know before a transaction closes.

Data preparation

The notebook profiles the original DLD export, reports missingness and duplicates, filters to Sales and Residential records, checks positive prices and plausible areas, standardises community names, and converts area from square metres to square feet.

Feature and leakage controls

The model uses area, parking, sale year/month, community, property type/subtype, room category, registration type and master project. It excludes actual_worth from predictors and excludes meter_sale_price because it is calculated using sale price and area. Identifier, rent, and Arabic duplicate fields are not used.

Model comparison

Two Linear Regression models use the same train/test split and feature set: one predicts raw AED prices; the second predicts log prices and converts predictions back with exp before evaluation. Compare R-squared, MAE, RMSE and MAPE on the original AED scale.

Business use and limitations

Use the model as a first-pass estimate with human review for luxury or unusual properties. Review errors by community and room category and compare with recent transactions. Historical associations are not causal; the dataset may omit property condition, view, floor, and other valuation drivers.

Important: Run the supplied notebook to generate dataset-specific metrics and plots. Verify every interpretation against the displayed output before submission. These materials are a working draft and must be reviewed.